

Sample Complexity Analysis for Identifying the Minimum Volume Enclosing Simplex

Kejun Huang

Department of Computer and Information Science and Engineering

University of Florida

Gainesville, Florida 32611

Email: kejun.huang@ufl.edu

Abstract—The minimum volume enclosing simplex (MVES) problem has found numerous applications in hyperspectral unmixing, topic modeling, community detection, and many other fields when data samples follow certain admixture structures in a latent domain. Its success not only thanks to natural representation power, but also the phenomenon that it is able to uniquely identify the latent components (up to permutation) when the data indeed follow a generative model with groundtruth latent components. A milestone result states that if the latent sources satisfy the assumption that they are sufficiently scattered in the probability simplex, then MVES is able to uniquely identify the latent components exactly. However, checking whether a given set of sources is NP-hard, which makes it difficult to quantify how often can we expect the method to be identifiable in practice. In this work, we provide a probabilistic analytical framework in attempt to answer this question. Assuming the latent sources are generated from a Sparse-Dirichlet model, we show that it will satisfy the sufficiently scattered condition with very high probability as long as $n \gg k \log(k)$, where k is the number of sources and n is the number of samples. This result is corroborated with synthetic experiments to validate that indeed the sufficiently scattered condition is a very mild condition to hold in practice.

Index Terms—simplex component analysis, minimum volume enclosing simplex, sufficiently scattered, identifiability.

I. INTRODUCTION

Given a collection of data points $\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \in \mathbb{R}^{k-1}$, which we stack them as columns of the matrix $\mathbf{X}$, the *minimum volume enclosing simplex* (MVES) problem aims at seeking, as the name suggests, the enclosing simplex with the minimum volume. A simplex in $\mathbb{R}^{k-1}$ is defined as the convex hull of k *affinely independent* vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ [1], i.e., $\{\mathbf{A}\mathbf{s} \mid \mathbf{s} \geq 0, \mathbf{I}^\top \mathbf{s} = 1\}$, where matrix $\mathbf{A}$ is obtained from stacking $\mathbf{a}_1, \dots, \mathbf{a}_k$ as its columns. If $\mathbf{A}$ defines an enclosing simplex for $\mathbf{X}$, then that means for every $\mathbf{x}_i$, there exists an $\mathbf{s}_i \geq 0$ and $\mathbf{I}^\top \mathbf{s}_i = 1$ such that $\mathbf{x}_i = \mathbf{A}\mathbf{s}_i$. Columns of $\mathbf{A}$ are called vertices of the simplex. It is easy to see that there is always an infinite number of enclosing simplexes, since we can always pull the vertices further away from the point clouds. Therefore, it makes sense to seek for the ‘smallest’ enclosing simplex, or more rigorously, the one with the minimum volume. The volume of the simplex defined by $\mathbf{A}$ has been shown

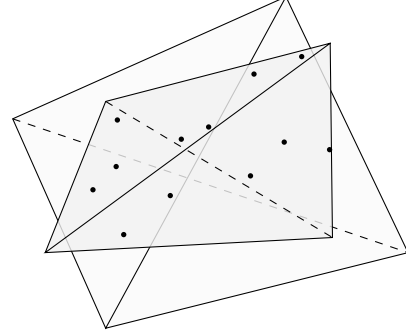


Fig. 1. A geometric interpretation of (1) of finding the minimum volume enclosing simplex for a set of points.

to be proportional to $\sqrt{\det(\mathbf{A}^\top \mathbf{A} + \mathbf{I}\mathbf{I}^\top)}$, which leads to the optimization problem

$$\begin{aligned} & \underset{\mathbf{A}, \mathbf{S}}{\text{minimize}} && \det(\mathbf{A}^\top \mathbf{A} + \mathbf{I}\mathbf{I}^\top) \\ & \text{subject to} && \mathbf{X} = \mathbf{A}\mathbf{S}, \quad \mathbf{S} \geq 0, \quad \mathbf{I}^\top \mathbf{S} = \mathbf{I}^\top. \end{aligned} \quad (1)$$

Notice that in practice the data points may lie on a subspace of a higher dimension $\mathbb{R}^m$, which is also acceptable for formulation (1). Figure 1 shows an illustration of MVES.

MVES is not only an interesting mathematical problem of its own, but has also found numerous applications in machine learning and signal processing, including hyperspectral unmixing [2], topic modeling [3], hidden Markov models identification [4], community detection [5], and crowdsourcing [6], to name just a few. As long as the data samples are assumed to be composed of *admixture* of several components/sources in an unsupervised manner, MVES can be used as the mathematical tool for data analytics.

One of the biggest benefit of using either formulation (1) in practice, other than its natural representation of admixture models, is their potential of identifying the correct latent factors under mild conditions. In a nutshell, if the groundtruth factor $\mathbf{S}$ satisfies a so-called *sufficiently scattered* condition in the probability simplex and $\mathbf{A}$ has independent columns, then the solutions of (1) must be the groundtruth $\mathbf{S}$ up to row permutation [7], [8]. Its identifiability is also highly related to some other structural constraints, such as nonnegative [9]–[11], bounded [12]–[14], and sparse [15]–[17].

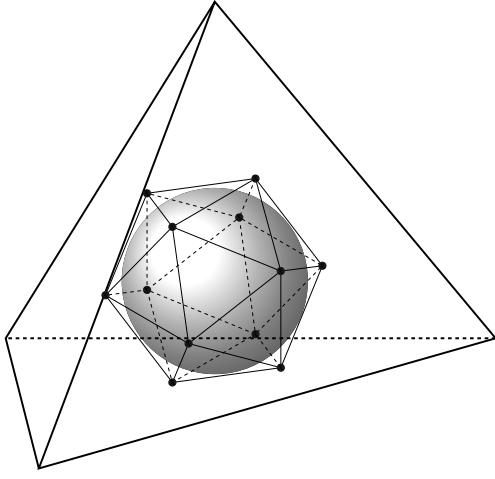


Fig. 2. A geometric interpretation of sufficiently scattered in the probability simplex in $\mathbb{R}^4$.

Assumption 1. Denote Δ as the probability simplex

$$\Delta = \{\mathbf{v} \in \mathbb{R}^k \mid \mathbf{v} \geq 0, \mathbf{I}^\top \mathbf{v} = 1\}$$

and $\bigcirc$ is the following Euclidean ball inside Δ

$$\bigcirc = \{\mathbf{v} \in \mathbb{R}^k \mid \|\mathbf{v}\|_2 \leq 1/\sqrt{k-1}, \mathbf{I}^\top \mathbf{v} = 1\},$$

a set $\mathcal{S}$ is sufficiently scattered in the probability simplex if:

- 1) $\bigcirc \subseteq \mathcal{S} \subseteq \Delta$;
- 2) $\partial \bigcirc \cap \partial \mathcal{S} = \{(1/(k-1))(\mathbf{I} - \mathbf{e}_c), c = 1, \dots, k\}$

where ∂ denotes the boundary of a set.

Theorem 1. Consider the admixture model $\mathbf{X} = \mathbf{A}_{\mathfrak{H}} \mathbf{S}_{\mathfrak{H}}$, where $\mathbf{A}_{\mathfrak{H}} \in \mathbb{R}^{d \times k}$ is the groundtruth mixing matrix and $\mathbf{S}_{\mathfrak{H}}$ is the groundtruth admixture coefficient matrix that satisfies $\mathbf{S}_{\mathfrak{H}} \geq 0$ and $\mathbf{I}^\top \mathbf{S}_{\mathfrak{H}} = \mathbf{I}^\top$. If columns of $\mathbf{A}$ are affinely independent and $\text{conv}(\mathbf{S}_{\mathfrak{H}})$ is sufficiently scattered in the probability simplex as in Assumption 1, then for any solution of (1), denoted as $(\mathbf{A}_\star, \mathbf{S}_\star)$, there exist a permutation matrix Π such that

$$\mathbf{A}_{\mathfrak{H}} = \mathbf{A}_\star \Pi \quad \text{and} \quad \mathbf{S}_{\mathfrak{H}} = \Pi^\top \mathbf{S}_\star.$$

In other words, MVES is identifiable if the groundtruth columns of $\mathbf{A}_{\mathfrak{H}}$ are affinely independent and the convex hull of $\mathbf{S}_{\mathfrak{H}}$ is sufficiently scattered in the probability simplex.

An illustration of sufficiently scattered in the probability simplex in $\mathbb{R}^4$ is shown in Figure 2. It is the most important breakthrough in the identifiability of MVES. However, this condition is also NP-hard to check exactly [18]. The goal of this paper is to show that if $\mathbf{S}_{\mathfrak{H}}$ is generated from a specific probabilistic model called sparse-Dirichlet, then its convex hull will be sufficiently scattered in the probability simplex as long as the number of samples n is much larger than $k \log(k)$, thus giving a more intuitive result that the sufficiently scattered condition is indeed a fairly mild condition to hold in practice.

II. SAMPLE COMPLEXITY ANALYSIS

Checking whether the convex hull of $\mathcal{S}$ satisfies Assumption 1 is NP-hard, as it has been shown to be equivalent to checking whether the solution set of the following nonconvex optimization problem is the set of unit vectors $\mathbf{e}_1, \dots, \mathbf{e}_k$:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} \quad \|\mathbf{w}\|_2^2 \\ & \text{subject to} \quad \mathbf{w}^\top \mathbf{S} \geq 0, \mathbf{w}^\top \mathbf{I} = 1. \end{aligned} \quad (2)$$

On the other hand, it has been pointed out that for $\text{conv}(\mathcal{S})$ to be sufficiently scattered in the probability simplex, it is necessary that every row of $\mathbf{S}$ contains at least $k-1$ zeros. This gives the intuition that a wide and sparse $\mathbf{S}$ is more likely to satisfy Assumption 1.

To this end, we propose the following generative model for $\mathbf{S}$. Since each column is a vector in the probability simplex, the obvious choice is to assume that they are generated from a Dirichlet distribution; however, since we also need to make sure that a significant portion of $\mathbf{S}$ are *exact* zeros, we propose to modify the more common Dirichlet distribution to be the so-called sparse-Dirichlet model. Readers may notice that some elements are first independently drawn from a gamma distribution before the columns are normalized to sum to one, which is a well-known way to generate a (original) Dirichlet random variable [19].

Assumption 2 (Sparse-Dirichlet model). The matrix $\mathbf{S} \in \mathbb{R}^{k \times n}$ is generated from a Sparse-Dirichlet model with parameter $p \in (0, 1)$, denoted as $\mathbf{S} \sim \mathcal{SD}(p)$, if its elements are i.i.d. with $S_{ij} = b_{ij} g_{ij} / h_i$, where $b_{ij} \in \{0, 1\}$ are i.i.d. Bernoulli random variables with $\Pr[b_{ij} = 1] = p$ (thus $\Pr[b_{ij} = 0] = 1 - p$), g_{ij} are i.i.d. random variables drawn from a gamma distribution with parameter $(\alpha, 1)$, and $h_i = \sum_j b_{ij} g_{ij}$ is the normalizing factor.

Although MVES requires the columns of $\mathbf{S}$ to sum to one, it turns out their scaling does not affect the optimization problem 2. Therefore, we will effectively assume that every element of $\mathbf{S}$ is i.i.d. with $S_{ij} = b_{ij} g_{ij}$ where b_{ij} are i.i.d. Bernoulli and g_{ij} are i.i.d. gamma.

If the optimal value of (2) is bigger than 1, then it means, equivalently, that there exists some point $\mathbf{w}$ in the $\{\mathbf{w} \mid \|\mathbf{w}\|_2 = 1, \mathbf{I}^\top \mathbf{w} = 1, \mathbf{w} \neq \mathbf{e}_1, \dots, \mathbf{e}_k\}$ such that $\mathbf{w}^\top \mathbf{s}_i \geq 0$ for all $i = 1, \dots, n$. In other words,

$$\Pr \left[\sup_{\substack{\mathbf{w}^\top \mathbf{S} \geq 0 \\ \mathbf{w}^\top \mathbf{I} = 1}} \|\mathbf{w}\|_2 > 1 \right] = \Pr \left[\sup_{\substack{\|\mathbf{w}\|_2 = 1 \\ \mathbf{w}^\top \mathbf{I} = 1}} \min_i \mathbf{w}^\top \mathbf{s}_i > 0 \right].$$

If the choices of $\mathbf{w}$ is finite, then one could easily bound right-hand-side probability by the union bound. However, the $(k-1)$ -sphere $\{\mathbf{w} \mid \|\mathbf{w}\|_2 = 1, \mathbf{I}^\top \mathbf{w} = 1\}$ clearly contains infinitely many points. To this end, we are going to use the following notion of δ -cover from convex geometry [18] that holds for all ℓ_p norm balls, but we are only to instantiate the Euclidean ball:

Lemma 1 (δ -cover). A finite δ -cover of the unit sphere in $\mathbb{R}^k$ is a finite set $\mathcal{C}_\delta$ of points with unit ℓ_2 norm such that any

point on the unit sphere is within ϵ away from an element in C_δ , i.e.

$$\min_{\mathbf{w}_i \in C_\delta} \|\mathbf{w} - \mathbf{w}_i\|_2 < \delta, \quad \forall \|\mathbf{w}\|_2 = 1.$$

For $\delta \in (0, 1)$ there always exists an δ -cover C_δ with cardinality $|C_\delta| < (3/\delta)^{k-1}$.

For the $(k-1)$ -sphere $\{\mathbf{w} \mid \|\mathbf{w}\|_2 = 1, \mathbf{I}^\top \mathbf{w} = 1\}$ that we are interested in, its radius is $\sqrt{(k-1)/k}$. Let $C_\delta = \{\mathbf{w}_j\}$ be an δ -cover for it, then its cardinality can be bounded as $|C_\delta| < (3(k-1)/(k\delta))^{k-1}$. Assume that we have both the lowerbound

$$\min_i \mathbf{s}_i^\top \mathbf{w}_j \geq \beta, \quad \forall \mathbf{w}_j \in C_\delta$$

and the upperbound

$$\max_i \|\mathbf{s}_i\|_2 \leq \gamma.$$

Then

$$\begin{aligned} \min_i \mathbf{s}_i^\top \mathbf{w} &\geq \min_i \mathbf{s}_i^\top \mathbf{w}_j + \min_i \mathbf{s}_i^\top (\mathbf{w} - \mathbf{w}_j) \\ &\geq \beta - \max_i \|\mathbf{s}_i\|_2 \delta \\ &\geq \beta - \gamma \delta. \end{aligned}$$

Therefore

$$\sup_{\substack{\|\mathbf{w}\|_2=1 \\ \mathbf{w}^\top \mathbf{I}=1}} \min_i \mathbf{w}^\top \mathbf{s}_i \geq \beta - \gamma \delta.$$

As a result, we have

$$\begin{aligned} &\Pr \left[\sup_{\substack{\|\mathbf{w}\|_2=1 \\ \mathbf{w}^\top \mathbf{I}=1}} \min_i \mathbf{w}^\top \mathbf{s}_i \geq \beta - \gamma \delta \right] \\ &\leq \sum_{\mathbf{w}_i \in C_\delta} \Pr \left[\min_i \mathbf{s}_i^\top \mathbf{w}_j \geq \beta \right] + \Pr \left[\max_i \|\mathbf{s}_i\|_2 \leq \gamma \right] \\ &= \sum_{\mathbf{w}_i \in C_\delta} \Pr \left[\mathbf{s}_i^\top \mathbf{w}_j \geq \beta \right]^n + \Pr \left[\|\mathbf{s}_i\|_2 \leq \gamma \right]^n, \end{aligned} \quad (3)$$

where C_δ is a δ -cover of the $(k-1)$ -sphere $\{\mathbf{w} \mid \|\mathbf{w}\|_2 = 1, \mathbf{I}^\top \mathbf{w} = 1\}$ with cardinality $|C_\delta| < (3(k-1)/(k\delta))^{k-1}$. The second line is obtained from the union bound, and the third line is because of the assumption that each $\mathbf{s}_i$ is independently and identically distributed.

We are now ready to present the main theorem about the sample complexity for identifiable MVES:

Theorem 2. Suppose $\mathbf{S} \in \mathbb{R}^{k \times n}$ is generated from the Sparse-Dirichlet model $\mathcal{SD}(p)$. Then

$$\Pr \left[\sup_{\substack{\mathbf{w}^\top \mathbf{S} \geq 0 \\ \mathbf{w}^\top \mathbf{I}=1}} \|\mathbf{w}\|_2 > 1 \right] \leq 2 \exp(k \log(k) - n\alpha), \quad (4)$$

where $\alpha = -\log \max(1 - (p/2)^k, (1+p)/2)$, which satisfies $\alpha \geq 0$ and $= 0$ when $p = 0$ or 1 .

Proof. We prove by bounding the two terms on the right-hand-side of (3).

a) *Bounding the first term of (3):* Consider the complement of $\Pr[\mathbf{s}_i^\top \mathbf{w}_j \geq \beta]$

$$\Pr[\mathbf{s}_i^\top \mathbf{w}_j \leq \beta] \geq \prod_{c=1}^k \Pr[s_{ic} w_{jc} \leq w_{jc} \beta].$$

The k terms on the right-hand-side can take three possible values:

$$\Pr[s_{ic} w_{jc} \leq w_{jc} \beta] = \begin{cases} 1 & s_{ic} = 0 \text{ or } w_{jc} = 0 \\ \Pr[g_{ic} \leq \beta] & s_{ic} \neq 0 \text{ and } w_{jc} > 0 \\ \Pr[g_{ic} \geq \beta] & s_{ic} \neq 0 \text{ and } w_{jc} < 0 \end{cases}$$

Define $c(\beta)$ as

$$c(\beta) = \max(\Pr[g_{ic} \leq \beta], \Pr[g_{ic} \geq \beta]), \quad (5)$$

which obviously satisfies $1/2 < c(\beta) < 1$ for any $\beta > 0$, we have

$$\Pr[\mathbf{s}_i^\top \mathbf{w}_j \geq \beta] \leq 1 - (pc(\beta))^k. \quad (6)$$

b) *Bounding the second term of (3):* Consider the complement of $\Pr[\|\mathbf{s}_i\|_2 \leq \gamma]$

$$\Pr[\|\mathbf{s}_i\|_2 \geq \gamma] \geq \Pr[s_{ic} \geq \gamma] = \begin{cases} 0 & s_{ic} = 0 \\ \Pr[g_{ic} \geq \gamma] & s_{ic} \neq 0 \end{cases}$$

Borrowing the definition of $c(\beta)$ in (5), we have

$$\Pr[\|\mathbf{s}_i\|_2 \leq \gamma] \leq 1 - (1-p)c(\gamma). \quad (7)$$

Now combining (3), (6), and (7), we have

$$\begin{aligned} &\Pr \left[\sup_{\substack{\|\mathbf{w}\|_2=1 \\ \mathbf{w}^\top \mathbf{I}=1}} \min_i \mathbf{w}^\top \mathbf{s}_i \geq \beta - \gamma \delta \right] \\ &\leq \left(\frac{3(k-1)}{k\delta} \right)^{k-1} \left(1 - (pc(\beta))^k \right)^n + (1 - (1-p)c(\gamma))^n \\ &\leq \left(\frac{3}{\delta} \right)^k \left(1 - (pc(\beta))^k \right)^n + (1 - (1-p)c(\gamma))^n. \end{aligned} \quad (8)$$

Setting $\gamma = \beta/\delta$, $\beta = 1$, and $\delta = 1/(3k)$, we get $c(\beta) = 1/2$ and $c(\gamma) > 1/2$, then (8) simplifies to

$$\begin{aligned} &\Pr \left[\sup_{\substack{\|\mathbf{w}\|_2=1 \\ \mathbf{w}^\top \mathbf{I}=1}} \min_i \mathbf{w}^\top \mathbf{s}_i \geq 0 \right] \\ &\leq k^k \left(1 - (p/2)^k \right)^n + ((1+p)/2)^n \\ &\leq 2 \exp(k \log(k) - n\alpha), \end{aligned}$$

where $\alpha = -\log \max(1 - (p/2)^k, (1+p)/2)$, which satisfies $\alpha \geq 0$ and $= 0$ when $p = 0$ or 1 . $\square$

Theorem 2 shows that if $\mathbf{S}$ is generated from a Sparse-Dirichlet model, then the probability that it does not satisfy the sufficiently scattered condition can be upperbounded by (4), which approaches zero at an exponential rate as $n \gg k \log(k)$ provided p is neither 0 or 1, which makes sense as one cannot expect $\mathbf{S}$ to be completely dense or all zeros. As a result, we

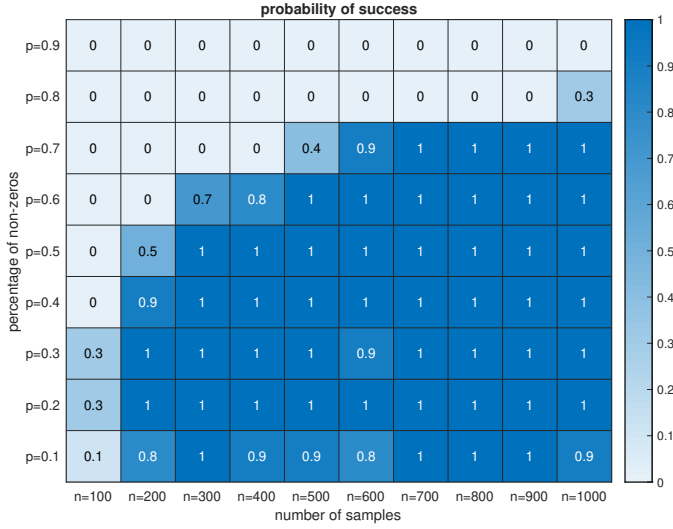


Fig. 3. Groundtruth $\mathbf{S}_h$ is generated from a Sparse-Dirichlet model. The probability of exactly recovering the dictionary for various sample size n and sparsity p (probability of nonzeros in $\mathbf{S}_h$). The size of the mixing matrix is fixed with $m = k = 20$.

have the following corollary that directly bounds the probability that such a Sparse-Dirichlet $\mathbf{S}$ can be uniquely identified via MVES.

Corollary 1. Consider the generative model $\mathbf{X} = \mathbf{A}_h \mathbf{S}_h$, where $\mathbf{A}_h \in \mathbb{R}^{m \times k}$ is the groundtruth mixing matrix and $\mathbf{S}$ is the groundtruth admixture coefficients. If columns of $\mathbf{A}_h$ are affinely independent and the matrix $\mathbf{S}_h \in \mathbb{R}^{k \times n}$ is generated from the Sparse-Dirichlet model $\mathcal{SD}(p)$, then $(\mathbf{A}_h, \mathbf{S}_h)$ are globally identifiable via optimizing (1) with probability at least

$$1 - 2 \exp(k \log(k) - n\alpha),$$

where $\alpha = -\log \max(1 - (p/2)^k, (1 + p)/2)$, which satisfies $\alpha \geq 0$ and $= 0$ when $p = 0$ or 1 .

III. EXPERIMENTS

We validate the sample complexity analysis of identifiable MVES in the following numerical experiment conducted in MATLAB. The algorithm to approximately solve formulation (1) is the L-ADMM algorithm proposed in [20], which takes the following form:

Algorithm 1 Solving MVES (1) by L-ADMM

- 1: $\mathbf{X} = \begin{bmatrix} \mathbf{I}^T \\ \mathbf{X} \end{bmatrix}$
 - 2: initialize $\mathbf{W}$ and $\mathbf{U}$
 - 3: $\rho = k$
 - 4: **repeat**
 - 5: $\mathbf{W} \leftarrow (\mathbf{S} + \mathbf{U} + (\mathbf{S}^\dagger)^\top / \rho) \mathbf{X}^\dagger$
 - 6: project columns of $\mathbf{W} \mathbf{X} - \mathbf{U}$ to Δ via bisection [21]
 - 7: $\mathbf{U} \leftarrow \mathbf{U} + \mathbf{S} - \mathbf{W} \mathbf{X}$
 - 8: **until** converge
-

The groundtruth latent components $\mathbf{S}_h$ is generated from a Sparse-Dirichlet model. We fix the mixing matrix of size $m = k = 20$, and change the sample size n from 100 to 1000 and the probability p in the Sparse-Dirichlet model from 0.1 to 0.9. We declare success if the estimation error (after resolving the permutation via the Hungarian algorithm [22]) is smaller than $1e-5$. In each setting the experiments are repeated 10 times and the percentage of exact recovery are recorded. The results are shown in Figure 3, which agrees with the bound in Theorem 2.

IV. CONCLUSION

In this work we present a novel analysis framework to answer the following question: how many samples are needed to guarantee that the minimum volume enclosing simplex exactly recovers the latent sources with high probability? We know a deterministic condition is that the sources are sufficiently scattered in the probability simplex, but unfortunately this condition is NP-hard to check. Assuming the latent sources are generated from a Sparse-Dirichlet model, we show that it will satisfy the sufficiently scattered condition with very high probability as long as $n \gg k$, where k is the number of sources and n is the number of samples. This result is corroborated with synthetic experiments to validate that indeed the sufficiently scattered condition is a very mild condition to hold in practice.

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